Will Humankind Survive the Zombie Apocalypse? Part II

Manufacturing Process Memorandum

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1 Introduction

1.1 Problem description

A biochemical reactor of volume $V = 25\,\text{m}^3$ is used to produce a vaccine against a zombie virus. Virus and chemical solutions are pumped in through two pipes, react inside the reactor, and the resulting vaccine is collected through a third pipe. The concentrations of virus c_Z, chemical c_C, and vaccine c_P inside the reactor are modelled by a system of three ODEs given in Eq. (1), derived from mass balance and the reaction knowledge (as done in Exercise class) $Z + C \to P$. The outflow rate satisfies $Q_3 = Q_1 + Q_2 = 5$ m^3/min by mass balance, ensuring constant reactor volume. The initial condition $c(0) = (0, 0, 0)^\top$ means that the reactor starts empty.

1.2 Goal

The goal is to identify which of the five candidate numerical methods among Explicit Euler, Improved Euler, AB2, AM2, and BDF2 is best suited to solve this system reliably. Given the inflow rates and concentrations, the goal is to model how c_Z, c_C, and c_P evolve over time, so that engineers can predict and optimize vaccine production under different operating conditions. Methods are compared in terms of accuracy, speed, and robustness. Robustness is tested by varying the virus inflow rate as $Q_{1,\text{new}} = \alpha Q_1$ with $\alpha \in [0.7, 1.3]$ and measuring how much the solution changes across methods. The biochemical reactor produces vaccine P via the reaction $Z + C \to P$. Denoting concentrations $c = (c_Z, c_C, c_P)^\top$, the ODE model reads

$$c' = f(t, c), \qquad f(t, c) = \begin{pmatrix} \dfrac{Q_1 c_{Z,0}}{V} - \dfrac{Q_3 c_Z}{V} - k\, c_Z c_C \\ \dfrac{Q_2 c_{C,0}}{V} - \dfrac{Q_3 c_C}{V} - k\, c_Z c_C \\ -\dfrac{Q_3 c_P}{V} + 2k\, c_Z c_C \end{pmatrix}, \tag{1}$$

with initial condition $c(0) = (0, 0, 0)^\top$ and parameters listed in Table 1.

Symbol	Value	Unit
V	25	m^3
Q_1	2.5	m^3/min
Q_2	2.5	m^3/min
$Q_3(Q1 + Q2)$	5	m^3/min
$c_{Z,0}$	20	mg/m^3
$c_{C,0}$	25	mg/m^3
k	1.2	m^3/(mg·min)

Table 1: Model parameters.

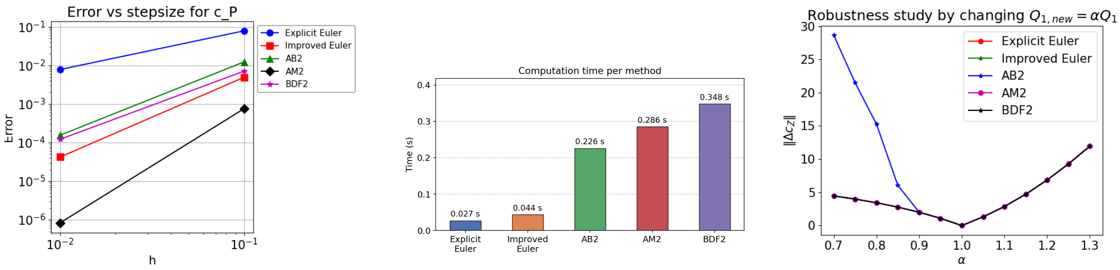
1.3 Model limitations

The model has the following limitations: it assumes perfect mixing inside the reactor, so concentrations are the same everywhere at all times. It also assumes the reactor is always completely full, so the volume V stays constant throughout. The reaction rate k is assumed constant, meaning any effects of temperature or pH are ignored. Finally, the model assumes steady and constant inflow rates and concentrations, so real-world fluctuations in the intake/outtake are not captured.

2 Scenarios

2.1 Accuracy

Figure 1a shows the global error in vaccine concentration (c_P) against step size h for all five methods. This is a log-log plot; the slope of each line gives the convergence order; a slope of



(a) Accuracy (b) Speed (c) Robustness

Figure 1: Comparison of numerical methods.

1 corresponds to first-order convergence, and a slope of 2 corresponds to second-order convergence. BDF2, Improved Euler, and AB2 achieve second-order convergence but with a slightly higher error than AM2. Explicit Euler is least accurate with first-order convergence.
Overall, AM2 achieves the lowest error across both step sizes, with second-order convergence clearly visible from the steep slope. We conclude, AM2 is the most accurate method overall for this problem.

2.2 Speed

Figure 1b shows the average computation time per method over 10 runs with $h = 0.01$. Explicit Euler is the fastest at 0.027 s, followed by Improved Euler at 0.044 s. Both are significantly faster than the two-step methods, as they require no Newton iteration and no startup procedure.. AB2 takes 0.226 s, AM2 0.286 s, and BDF2 is the slowest at 0.348 s. The higher cost of AM2 and BDF2 is expected since they solve a nonlinear system at every step via Newton iteration. Computation times were measured using `timeit` with 10 runs at $h = 0.01$, on a system with an Intel Core i5-12500H (12 cores, 16 threads) and 15 GiB RAM, running Fedora 44 (kernel 7.0.10, x86_64).

2.3 Robustness

Figure 1c shows how much the solution changes in c_Z when the virus inflow rate is varied as $Q_{1,\text{new}} = \alpha Q_1$ with $\alpha \in [0.7, 1.3]$ and step size $h = 0.2$. For each method, $\|\Delta c_Z\|$ denotes the norm of the difference between its baseline solution ($\alpha = 1$) and its perturbed solution at each α. A robust solution method should closely track the true change in c_z. A non-robust method either overreacts or underreacts to the perturbation, meaning you cannot trust it when the operating conditions change slightly. All methods show zero deviation at $\alpha = 1.0$ by definition, since that is the baseline. AB2 is clearly the least robust, showing very large deviations especially for $\alpha < 1.0$, which is consistent with its instability at larger step sizes. Explicit Euler, Improved Euler, AM2 and BDF2 all behave similarly and remain close to each other across the full range of α, indicating good robustness.

3 Recommendation

The most important criteria for this problem are accuracy and robustness as the fate of humanity depends on this, a wrong answer is far far worse than a slow one. Both AM2 and BDF2 are A-stable for $k = 2$, meaning their stability regions contain the entire left complex half-plane [1, Sec. 6.2]. By the Second Dahlquist Barrier [1, Theorem 98], AM2 (the trapezoidal rule) is the A-stable second-order method with the smallest error constant among all A-stable linear multistep methods, which explains its superior accuracy over BDF2 in Figure 1a. This theoretical guarantee, combined with the empirical results, makes AM2 the recommended method for accuracy.
In addition AM2 is as robust as the other methods as seen in Figure 1c, while not the fastest, its increased computation time is justifiable due to the gain in accuracy. Overall, AM2 is the recommended method for this problem

References

[1] C. Seifert, D. Ruprecht, J. Urizarna Carasa, *Numerical Methods for ODEs – Lecture Script*, TU Hamburg, 2023.